

Blocco 3

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In the following "hf" will stand for hyperfinite and hhf will stand for hyperhyperfinite.

Esercizio 1. Define an equivalence relation on the Baire space by setting xEy iff for all large n , $x_n = y_n$. Prove that E is hf.

Proof.

□

Esercizio 2. Show that $E \leq_B E'$ and E' is hhf, then E is also hhf.

Proof. Since E' is hf, we can write $E' = \bigcup_{n < \omega} E'_n$ with $\langle E'_n : n < \omega \rangle$ increasing sequence of hf cbers. Let $f : X \rightarrow X$ be a Borel reduction from E to E' . For every $n < \omega$ define $xE_ny \iff f(x)E'_nf(y)$.

- $E \subseteq \bigcup_{n < \omega} E_n$:

$$xEy \Rightarrow f(x)E'f(y) \Rightarrow f(x)E'_nf(y) \text{ for some } n < \omega \Rightarrow xE_ny \Rightarrow x \bigcup_{n < \omega} E_ny \quad (1)$$

- $\bigcup_{n < \omega} E_n \subseteq E$:

$$x \bigcup_{n < \omega} E_n \Rightarrow xE_ny \text{ for some } n < \omega \Rightarrow f(x)E'_nf(y) \text{ for some } n < \omega \Rightarrow f(x)E'f(y) \Rightarrow xEy \quad (2)$$

- for all $n < \omega$ E_n is hf cber :

Fix $n < \omega$. Then E_n is a cber because f, E'_n are Borel and $E_n \subseteq E$. E_n is hf because by definition f is a Borel reduction from E_n to E'_n and E'_n is Borel.

- $\langle E'_n : n < \omega \rangle$ is increasing :

In fact $\langle E'_n : n < \omega \rangle$ is increasing, so we get

$$xE_ny \Rightarrow f(x)E'_nf(y) \Rightarrow f(x)E'_{n+1}f(y) \Rightarrow f(x)E'f(y) \Rightarrow xEy \quad (3)$$

We conclude observing that $E = \bigcup_{n < \omega} E_n$ witnesses E hf.

□

Esercizio 3. Show that the Cohen forcing $Add(\omega, 1)$ is weakly homogeneous.

Proof.

□

For the next problem, let k be a measurable cardinal with a normal measure U and let P be the Prickry forcing at k .

Esercizio 4. Prove that P is weakly homogeneous.

Proof.

□